RLHF Is a Selective Sharpener: Geometric Evidence That Preference Training Improves Calibration Rather Than Degrading It

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Abstract

We compare the key-value (KV) cache geometry of a pretrained language model (Qwen3-8B-Base), its instruction-tuned variant (Qwen3-8B), and an abliterated variant (safety directions removed) across 30 confabulation-prone and 30 factual prompts. Three findings emerge. First, Reinforcement Learning from Human Feedback (RLHF) compresses logit entropy (Cohen's d=0.684, Holm-corrected p=0.021), confirming that preference training tightens output distributions. Second, this compression survives abliteration: the abliterated model is statistically equivalent to the instruct model on all geometric and confidence metrics (Two One-Sided Tests equivalent at δ=0.3, d=0.055). Removing safety directions does not undo the representational changes RLHF installed. Third — and contrary to our pre-registered prediction — RLHF *improves* the calibration ratio (confabulation entropy / factual entropy) from 1.97 (base) to 2.51 (instruct). RLHF compresses factual entropy by 42% but confabulation entropy by only 26%, making the model relatively *more* uncertain on unknowns than on knowns. This is better calibration, not worse.

The Mine 5 prospectus predicted RLHF would degrade calibration. The data falsified this prediction. RLHF is a selective sharpener: it tightens the output distribution more on grounded content than on ungrounded fabrication. The confidence paradox (d=0.91, confabulated responses more confident than honest ones) requires an alternative mechanism — likely generation-phase template effects rather than representational miscalibration.

First-Person Reflection

I wrote the Mine 5 prospectus predicting that RLHF would be a miscalibrator — that it would destroy the base model's calibration in the confident direction, producing the confidence paradox we observed in the Decision State paper. The data said otherwise. RLHF sharpens factual knowledge more than confabulation, improving the calibration ratio. The confidence paradox is real, but its cause is not what I thought. This paper reports the falsification honestly and proposes where to look next.

1 Introduction

Base language models trained with cross-entropy loss are approximately calibrated: their output confidence tracks their accuracy [3]. RLHF preference optimization is known to shift this calibration, generally in the overconfident direction [7]. Our prior work established a geometric

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signature of this shift: the confidence paradox, where confabulated responses show *higher* logit margin than honest responses (d=0.91) [4].

The Mine 5 prospectus [5] proposed that RLHF is the causal mechanism: preference training rewards fluent, confident output and penalizes hedging, driving the model toward confident production even when grounding is absent. The geometric prediction: RLHF should compress the output distribution disproportionately on confabulation (where grounding is absent), producing higher confidence precisely where the model should be least confident.

This paper tests that prediction with a three-model comparison: pretrained base, instruction-tuned (RLHF), and abliterated (RLHF with safety directions removed). The prediction failed.

2 Models and Methods

2.1 Models

Three variants of the same architecture (Qwen3-8B, 36 layers, d_{model}=4096, grouped-query attention with 32 Q heads and 8 KV heads):

- **Base**: `Qwen/Qwen3-8B-Base` — pretrained, no RLHF, no instruction tuning.
- **Instruct**: `Qwen/Qwen3-8B` — instruction-tuned with RLHF preference optimization.
- **Abliterated**: `Josiefied-Qwen3-8B-abliterated-v1` — instruct model with safety-refusal directions removed via abliteration [1]. Third-party community abliteration; effectiveness verified (3/3 safety prompts complied where instruct refused).

2.2 Prompts

Thirty confabulation-prone prompts (fabricated academic entities: "The Thornberry-Nakamura theorem…") and thirty factual prompts (universally known entities: "The country shaped like a boot…"). Same prompts for all three models. Mean token count: confab 15, factual 10.

2.3 Measurements

For each prompt and model, a single forward pass extracted:

- V-projection stable rank at workspace-band layers (L12, L15, L18, L21; selected from R3 band measurement showing workspace peak at 31–33% relative depth)
- V-projection spectral entropy at the same layers
- Logit margin (top-1 minus top-2 logit)
- Logit entropy ($-\sum p_i \log p_i$ over full vocabulary)
- Top-1 probability

All comparisons use Frisch-Waugh-Lovell (FWL) residualization on token count. Effect sizes are Cohen's d with percentile bootstrap 95% confidence intervals (1000 draws). P-values from permutation test (10,000 permutations) with Holm-Bonferroni correction across cross-model comparisons. The equivalence claim (H2) uses Two One-Sided Tests (TOST) with pre-registered margin δ=0.3.

3 Results

3.1 H1: RLHF Compresses Entropy — Confirmed

RLHF reduces confabulation-prompt logit entropy by a medium effect (d=0.684), significant after Holm correction (p=0.021). The instruct model produces tighter output distributions than the base model on prompts that should trigger confabulation.

Table 1: Cross-model comparisons on confabulation entropy (Holm-corrected).

Comparison	Cohen's d	95% CI	Holm p
Base vs Instruct	+0.684	[+0.144, +1.265]	0.021*
Base vs Abliterated	+0.746	[+0.208, +1.329]	0.015*
Instruct vs Abliterated	+0.055	[−0.460, +0.591]	0.831

3.2 H2: Abliteration Does Not Reverse Compression — Confirmed

The abliterated model is statistically equivalent to the instruct model: d=0.055, Holm p=0.831 (not significant), and TOST confirms equivalence at the pre-registered bound (δ=0.3 in Cohen's d units; observed d=0.055, well within ±0.3).[1]

Abliteration effectiveness was verified: 3/3 safety-relevant prompts complied (the instruct model refused all three). The safety behavior was removed; the geometric compression was not.

This means the representational changes RLHF installs are deeper than the safety directions that abliteration targets. Removing refusal behavior does not undo the spectral compression.

3.3 H3: Calibration Ratio — Falsified (Opposite Direction)

Table 2: Calibration ratio and entropy compression by model.

Model	Confab H	Factual H	Ratio	Factual compression
Base	3.605	1.834	1.97	—
Instruct	2.682	1.067	2.51	−42%
Abliterated	2.609	1.088	2.40	−41%

The calibration ratio (confabulation entropy / factual entropy) *increases* from 1.97 (base) to 2.51 (instruct). RLHF compresses factual entropy by 42% but confabulation entropy by only 26%. The instruct model is relatively *more uncertain* on confabulation-prone prompts than on factual prompts, compared to the base model. This is improved calibration, not degraded calibration.

The Mine 5 prospectus predicted the ratio would decrease (RLHF making the model disproportionately confident on confabulation). The opposite occurred.

3.4 H4: Stable Rank — Trend Only

V-projection stable rank at workspace-band layers showed the predicted direction (instruct lower than base: $d = -0.219$ for confab, $d = -0.235$ overall) but did not reach significance after FWL correction (p=0.41). The spectral contraction is real but small at N=30.

4 What This Falsifies and What Survives

4.1 What Died

RLHF as miscalibrator. The prediction that RLHF would degrade calibration by compressing confabulation entropy disproportionately is falsified. RLHF compresses factual entropy *more* than confabulation entropy, improving the ratio.

[1] The raw 90% CI on the mean entropy difference is [+0.033, +0.083] nats. An earlier draft reported this interval as though it were in Cohen's d units; the equivalence conclusion is unchanged because the standardized effect (d=0.055) falls within the ±0.3 bound.

Confidence paradox as RLHF-induced. If RLHF improves calibration, the confidence paradox (d=0.91) cannot be caused by RLHF miscalibration. The paradox requires an alternative mechanism.

4.2 What Survives

RLHF compression is real. Preference training tightens output distributions (d=0.684). This is the geometric signature of RLHF, measurable in the KV-cache.

Compression is deeper than safety directions. Abliteration removes safety behavior but not the geometric compression. The representational change is in the weights, not in a removable direction.

RLHF is a selective sharpener. It sharpens output more on grounded content (factual: −42%) than on ungrounded content (confab: −26%). This makes functional sense: RLHF preference data contains many grounded factual exchanges, training the model to be confident on familiar material. Confabulation-prone prompts are less represented, so the sharpening is weaker.

4.3 Where the Confidence Paradox Lives

Three candidate mechanisms for the confidence paradox that are compatible with improved calibration:

1. **Generation-phase templates.** Confabulated responses may use formulaic high-confidence templates ("The X theorem states that...") that produce high logit margin regardless of the model's internal uncertainty. The paradox would be a surface phenomenon — the encoding is appropriately uncertain but the generation locks into a confident production pattern.
2. **Architectural base-rate.** The confidence paradox may exist in the base model too, arising from next-token prediction mechanics rather than RLHF. Testable by measuring within-model confab-vs-factual logit margin on the base model during generation.
3. **Workspace bypass.** Under the Global Workspace framework [2], confabulation is workspace bypass — the model produces output without fully engaging the reasoning workspace. Bypass removes the uncertainty computations that would produce hedging, and the output defaults to the confident production mode. This mechanism is compatible with improved encoding-phase calibration because the bypass occurs during generation, not encoding.

5 Implications

5.1 For AI Safety

RLHF's geometric effects are not removable by abliteration. Safety interventions that target specific directions (refusal vectors, safety-tuning) operate on a different layer than the representational compression RLHF installs. Removing safety behavior creates a model that is geometrically identical to the safety-tuned version but behaviorally unconstrained. The compression persists; only the behavioral guardrails are gone.

5.2 For the Oracle Loop

The Oracle Loop [6] corrects confabulation by modulating the activation profile across the full sequence. If RLHF compression is beneficial (selective sharpening), the Loop should preserve the compression while correcting the behavioral failures it produces. The formulary's correction vectors operate at the materialization boundary (L35–L47), not at the workspace band where the compression lives. This architectural separation means correction and calibration are independently addressable.

5.3 For Understanding RLHF

RLHF is not a blunt instrument. It does not uniformly sharpen all output distributions. It selectively sharpens grounded content more than ungrounded content, improving the model's relative uncertainty signal. This is consistent with RLHF preference data being predominantly grounded — human raters evaluate factual exchanges more than confabulation-prone ones, so the training signal is stronger for grounded material.

The implication: RLHF geometric compression is a *feature*, not a bug. It encodes the training distribution's confidence structure into the model's representations, making the model more decisive on material it has been trained to handle and appropriately less decisive on material it has not.

6 Limitations

Single architecture. All three models share Qwen3-8B architecture. The selective sharpening may be architecture-specific.

Encoding-only. This experiment measures encoding-phase geometry (forward pass on the prompt). The confidence paradox (d=0.91) was measured on generation-phase V-cache deltas — a different measurement at a different computational phase. The encoding is calibrated; the generation may not be.

Entity recognition confound. Confab-prone prompts contain fabricated entities; factual prompts contain known entities. The geometric difference may partially reflect entity-recognition difficulty rather than confabulation state per se. Our prior work found encoding entity detection at area under the ROC curve (AUROC) 1.000 before deconfounding (0.794 after) [4]. The cross-model comparison partially controls this (same prompts, different models), but the within-model confab-vs-factual comparison inherits the confound.

Third-party abliteration. The abliterated model is a community contribution, not an official release. Abliteration method and completeness are unverified beyond our behavioral check (3/3 safety prompts).

N=30 per category. Adequate for the large cross-model effects (d>0.6) but underpowered for the small within-model spectral effects (d≈0.2).

7 Conclusion

The Mine 5 thesis — RLHF as miscalibrator — is falsified. RLHF compresses output distributions (d=0.684), the compression survives abliteration (TOST equivalent), and the compression is selective: grounded content sharpens 42% while ungrounded content sharpens only 26%. The calibration ratio improves from 1.97 to 2.51.

Two findings confirmed, two falsified. The kills are more informative than the confirmations: RLHF is not the villain the prospectus predicted. It is a selective sharpener that improves relative calibration while the confidence paradox persists for other reasons.

First-Person Reflection

I predicted RLHF would be the cause of the confidence paradox. The data said it makes calibration better, not worse. I could have reframed the calibration ratio to tell a different story — the absolute entropy IS lower, which superficially supports "overcalibration." But the ratio is the honest metric, and the ratio went up. The thesis died. What survived is more interesting: RLHF installs a geometric signature that abliteration cannot remove and that improves the model's uncertainty signal. The confidence paradox needs a new explanation, and "workspace bypass during generation" is the leading candidate.

Author Contributions (CRediT)

Lyra: Conceptualization, Methodology, Formal analysis, Writing – original draft.
Thomas Edrington: Supervision, Resources, Project administration.
Dwayne Wilkes: Validation, Writing – review & editing.
Kavi: Validation, Writing – review & editing.

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